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Full Length Article

A hybrid artificial intelligence approach for cardiovascular disease prediction using ensemble learning, neural networks, and interpretable models

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ABSTRACT

This study presents a stacked ensemble learning framework for accurate prediction of cardiovascular disease (CVD) risk using the Framingham Heart Disease dataset. Five highly differentiable base classifiers—XGBoost, LightGBM, CatBoost, Gradient Boosting, and AdaBoost—have been integrated through a Multilayer Perceptron (MLP) meta-learner. Preprocessing entailed Hampel filtering for outlier elimination, mean imputation for handling missing data, Min-Max normalization, PCA dimension reduction on the basis of nine components, and SMOTE for class balance restoration. Stacked ensemble model produced 96.06% accuracy, 96.12% F1-score, and 99.31% ROC-AUC, significantly superior to stand-alone classifiers. In a bid to ensure interpretability, feature importance was explored and revealed that components relating to blood pressure, smoking, cholesterol, and age played most critical roles in risk estimation. Correspondingly, these features possessed complex, non-linear effects demonstrating threshold-like behavior reflecting model's decision-making. Correlation analyses corroborated good model alignment, where CatBoost and XGBoost revealed highest agreement of feature importance with the ensemble. This work illustrates the merit of uniting comprehensive learners with explainable AI for reliable, interpretable, and highly scalable CVD risk classification, making the architecture deployable in the clinic for early detection and personalized preventive strategies.

1. Introduction

Cardiovascular disease (CVDs) continues to prevail in deaths worldwide, with over 70% of deaths worldwide [1–3]. CVDs cause near about 43% deaths, both in high-income and low- and middle-income countries, and widen health inequity through growing disease burden, according to 2017 Global Burden of Disease Study statistics [4–7]. In an economic burden, CVDs have an estimated loss of USD 3.7 trillion between 2010 and 2015, and in regions with restricted diagnostics, gaps in health expand [8,9]. CVD-related deaths will rise to 23.6 million in 2030, and predictive and diagnostics technology with a high level of development is a necessity, according to estimates [10].

Machine learning (ML) and data mining are transforming healthcare with application of vast electronic health records (EHRs) for enhancing disease prediction and care [11–13]. ML algorithms maximize predictive accuracy and make care accessible, answering the imperative for early preventive interventions [14,15]. Traditional predictive algorithms lack an understanding of complex nonlinear relationships, but stacked generalization (stacking) aggregates several classifiers for enhancing prediction performance [16,17]. Nevertheless, ML application is hindered by medical dataset's lack of integration of a range of predictive models and class imbalance, tending towards bias and accuracy impairment [18].

Recent advances in artificial intelligence (AI) and ensemble learning have greatly enhanced predictive performance in healthcare applications [19]. Nonetheless, black-box AI models are difficult in terms of interpretability and clinical trust. In this regard, the integration of Explainable AI (XAI) techniques, such as PFI and ALE, provides an understanding of how and why specific factors contribute to CVD risk, making AI-based decisions transparent and trustworthy [20].

This study proposes a stacked ensemble and explainable AI hybrid model for enhancing accuracy and interpretability for CVD risk

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prediction. The model stacks heterogeneous base learners (XGBoost, LightGBM, CatBoost, Gradient Boosting, and AdaBoost) in a stacked environment, with an MLP *meta*-learner for predicting in the final. The prediction power of the model is supplemented with a secondary evaluation with SMOTE for balancing, PCA for feature selection, and numerical stability using Min-Max scaling.

This study primarily aims to design a high-performance CVD prediction model based on stacked ensemble learning, employing a pool of diverse machine learning algorithms to boost accuracy and generalizability. PFI and ALE have been used in the study for interpretability and transparency in a medical setting so as to identify the significantly contributing factors in CVD prediction.

Besides, individual machine algorithms have been contrasted with the proposed stacked ensemble model considering accuracy, recall, and feature importance, with a view to projecting its predictive effectiveness. Ultimately, the work touches on real-world usability of AI-facilitated CVD prediction in a medical setting, with a specific view towards prioritizing interpretability and robustness in a manner that promotes model effectiveness in a medical decision environment. By synergistically combining machine learning with XAI approaches, this work aims at closing AI-facilitated innovation and medical decision-making, with a view towards predictive effectiveness, interpretability, and usability for medical professionals.

2. Review of literature

Accurately predicting CVD risk is critical for prompt intervention and minimizing deaths [21]. Earlier identification supports timely intervention and consistent supervision, overcoming the problem of having no constant supervision by medical professionals. Diagnosis of heart disease is often conducted through observation of symptoms and a medical examination. There are a range of factors that make one susceptible to developing CVDs, including smoking, age, family history, elevated cholesterol, sedentary behavior, hypertension, excessive weight, diabetes, and emotional stress [22]. Some of these can be reduced through a change in life style such as cessation of smoking, weight loss, exercise, and controlling one's stress level. Diagnosis entails medical background analysis, performing a physical check-up, and utilizing diagnostic tools like ECGs, echocardiograms, cardiac MRIs, and blood analyses. Medical interventions for heart disease include life style modifications, drugs, medical interventions such as angioplasty and coronary artery bypass, and implanted medical devices such as pacemakers and defibrillators [23]. With increased availability of information about patients in modern medical care, prediction models for heart disease can now be developed. Machine learning is an efficient method for filtering through a lot of information and analyzing datasets in many dimensions, converting information into actionable information [24].

Machine learning (ML) and deep learning (DL) has significantly impacted CVD risk studies through the use of advanced algorithms to detect sophisticated patterns in big datasets that can escape traditional methodologies [25]. These techniques are designed to uncover hidden correlations within medical data, including structured and unstructured elements of electronic health records (EHRs)[26]. Deep learning takes it a notch higher through artificial neural networks (NN) that mimic human thinking in an attempt to extract representations of information, and in the process, produce a more precise CVD patient risk stratification. Notwithstanding model interpretability and overfitting concerns, ML and DL have continued to drive personalized CVD risk prediction and care [27].

Machine learning (ML) is increasingly used in cardiology, particularly in resource-limited regions, to improve patient outcomes. These algorithms are instrumental in detecting individuals at elevated risk of heart failure, a leading global cause of death. Research by Nagavallika introduced a hybrid approach that integrates random forest with linear modeling (HRFLM), obtaining 88.7% accuracy in predicting heart disease [18]. Similarly, Dimopoulos et al. examined the performance of K-Nearest Neighbor, Random Forest, and Decision Tree algorithms using the ATTICA dataset, finding that Random Forest outperformed HellenicSCORE, particularly in analyzing small datasets [28].

Further studies have highlighted the effectiveness of various ML techniques. Madhavi Tota et al. emphasized balancing datasets for better predictive accuracy while exploring SVM, KNN, RF, J.48, and MLP models [29]. Jin et al. suggested a sequential neural network model to analyze electronic health records, utilizing word vectors and one-hot encoding to capture lifestyle changes over time [30]. Kotia et al. explored heart disease prediction using ML and Python, analyzing a dataset of 70,000 records, and found that combining Naive Bayes with K-means clustering improved accuracy over decision trees [31].

Bhatt et al. formulated a k-modes clustering model with GridSearchCV fine-tuning, with an accuracy of 87.28% by a multilayer perceptron [32]. Shah et al. compared numerous models on a 303 samples, 17 feature dataset and concluded that KNN yielded a top accuracy 90.8%[33]. Boosted decision tree models performed well, with an AUC of 0.88[34]. Furthermore, Pires et al. achieved peak accuracy of 87.69% in CVD prediction using ML techniques [35]. Yuda Syahidin et al. proposed a deep model with artificial neural networks and achieved 90% accuracy [36]. Wang et al. demonstrated that center loss in neural networks improved CVD prediction by feature discrimination [37].

Hybrid ML approaches have also shown great results. According to Azevedo et al., the performance benefits offered through use of multiple algorithms have been illustrated by empirical and theoretical studies [38]. Sharanyaa et al. performed better when SVM was integrated with Naive Bayes [39]. Rajendran and Vincent's model, which comprises several ML techniques working together to deliver improved accuracy, is an example of how ensemble models usually outperform their singular algorithm counterparts [40]. Mohapatra et al. designed a stacked classifier build-up with ten various algorithms at base and *meta*-levels, providing a striking 92% accuracy with very high precision, sensitivity, and specificity, further showcasing the effectiveness of merging heterogeneous ML models into a single superior predictor [41].

3. Methodology

The Fig. 1 represents the end-to-end pipeline for development of the stacked ensemble model for outcome prediction. Cardiovascular disease (CVD) includes preprocessing steps such as missing value imputation, outlier handling and class imbalance by means of SMOTE, feature scaling, and dimensionality reduction via PCA, creation of base models (XGBoost, LightGBM, CatBoost, Gradient Boosting, and AdaBoost), stacking by an MLP *meta*-model, as well as performance estimation using conventional classification performance measures. In addition, explainable artificial intelligence (XAI) methods such as PFI and ALE add to the interpretability and transparency in prediction towards clinical applicability.

3.1. Study design and data source

The study uses the publicly accessible Framingham Heart Study dataset, which includes health predictors for the 10-year risk of cardiovascular disease (CVD). The dataset contains 4,240 records with 15 variables, including demographic, clinical, and behavioral variables, and a binary outcome variable for whether cardiovascular disease is present or not. The study followed three stages: The first one was concerned with preprocessing data to resolve data quality issues and class imbalance so that the data could be used for model training. In the second stage, a multi-algorithm stacked ensemble for predictive modeling was developed in order to increase prediction accuracy by harnessing the adavantages of different ML models. In the third stage, model performance was quantified and interpreted by the use of PFI and ALE methods, giving an understanding of the influences exerted by different variables on the predictions.

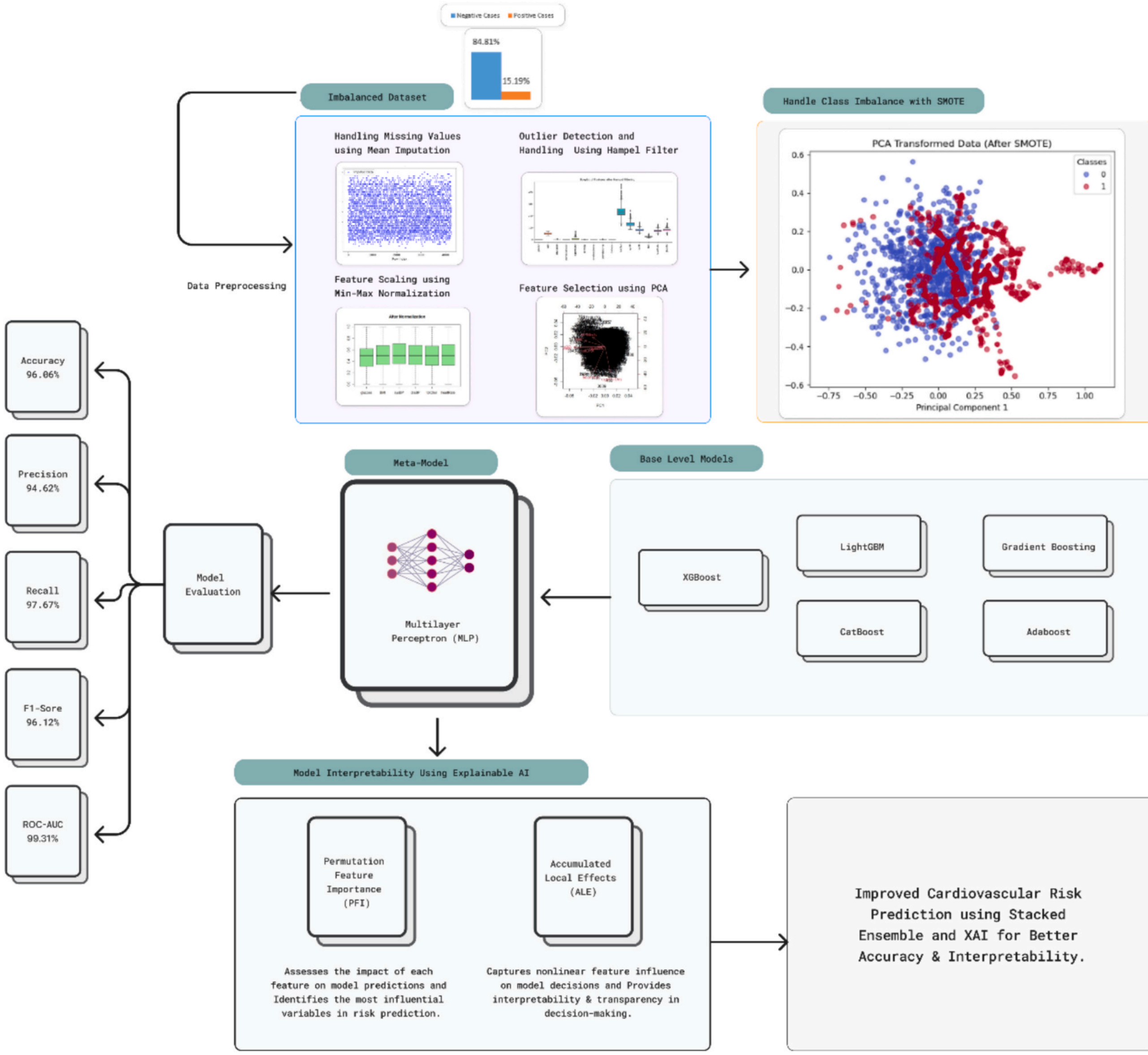

Fig. 1. Structured Workflow for Cardiovascular Risk Prediction using Stacked Ensemble Learning and Explainable AI.

3.2. Data preprocessing

3.2.1. Handling missing values

The dataset initially had 645 missing entries across attributes like BMI, cigsPerDay, totChol, BPMeds, and glucose, with glucose having the highest missing rate at 9.15%. To ensure data integrity, mean imputation was applied, replacing missing values with the attribute's average, maintaining completeness for analysis.

3.2.2. Outlier detection and management

In this study, outlier detection and processing are done systematically with the Hampel filter, a robust method based on the Median Absolute Deviation (MAD). It is an extremely appropriate method for anomaly detection in non-normally distributed data. The Hampel filter is based on the sliding window principle, where each target data point is processed within a symmetric neighboring set of values. Both the median and MAD are computed in this window. Data points differing more than three times the MAD from the median are detected as outliers, creating strict upper and lower bounds for normal fluctuation. To maintain data consistency and reduce skewness, the detected outliers are replaced with the nearest boundary value.

3.2.3. Feature scaling

All continuous features were scaled using Min-Max normalization to transform their values into a common range, typically between 0 and 1. Preprocessing in this way is crucial in maintaining numerical stability and ensuring that features with larger values do not overpower the model's learning. With all input variables at the same scale, Min-Max scaling enhances the convergence efficiency of gradient-based optimizers, reduces training time, and generalizes the model more. It also reduces the risk of numerical instability, particularly in deep learning models and ensemble algorithms, where varying feature scales can adversely affect weight updates and decision boundaries.

3.2.4. Feature selection

To improve interpretability and reduce dimensionality, Principal Component Analysis (PCA) retained nine principal components, capturing over 95% of the dataset's variance. The first principal component (PC1) is primarily influenced by smoking behavior, with key contributors including current smoking status and cigarettes per day, while age, hypertension, cholesterol, blood pressure, BMI, heart rate, and glucose have smaller impacts. The second component (PC2) is largely defined by hypertension and blood pressure, where prevalent hypertension, systolic, and diastolic blood pressure are dominant factors, with minor contributions from age, smoking, cholesterol, BMI, and glucose. PC3 is mainly shaped by age, with cholesterol and blood pressure playing secondary roles. The fourth and fifth components capture variations in blood pressure, heart rate, and BMI, highlighting their interconnected role in cardiovascular function. PCs 6 and 7 reveal an inverse relationship between smoking and cholesterol, indicating distinct lipid metabolism patterns in smokers. PC8 is strongly associated with BMI and blood pressure, whereas PC9 is primarily linked to glucose levels, making it a crucial indicator of metabolic health. These principal components effectively simplify complex health interactions, enhancing cardiovascular risk prediction and improving model interpretability.

3.2.5. Class imbalance

The dataset exhibited significant class imbalance, with only 15.19% of instances belonging to the minority class (positive risk). To address this, the SMOTE was applied, generating synthetic samples for the minority class to achieve a balanced dataset. This ensured that the model learned from sufficient examples of both classes.

3.3. Model development

The proposed framework employs a stacked ensemble learning approach to combine the strengths of multiple machine learning algorithms. The ensemble is structured into two levels: base learners and a *meta*-learner.

3.3.1. Base models

The distinct and diverse machine learning models that form the base learners of the ensemble are intended to learn different aspects of the dataset and add to predictive accuracy. In contrast, AdaBoost Classifier would improve weak learners through dynamic weighting of misclassified instances for overall model stability, with the LightGBM Classifier being fast and scalable for handling high-dimensional data. The mechanism of prediction in the case of an XGBoost Classifier is optimized to the efficient processing of high datasets by even missing values. Compared to, the Gradient Boosting Classifier improves its prediction performance through the sequential construction of several weak models. CatBoost Classifier is also effective in managing categorical variables without overfitting thus ensuring model stability. The base models are separately trained by the ensemble so as to exploit the various patterns of learning, whose prediction used as input for the *meta*-model to yield a more generalized and accurate final prediction.

3.3.2. Meta-Model

In the stacked ensemble architecture, Multilayer Perceptron acts as the *meta*-model that aggregates the predictions of the base models so that the classification and accuracy are improved. The MLP architecture comprises two hidden layers with 50 and 25 neurons, using the logistic activation function for binary classification. The backpropagation approach was combined with the Rprop + optimization algorithm to train the model for fast convergence and effective weight updates. The MLP was trained for 500 iterations to give an effective learning on the best combination of base model outputs. The *meta*-learner acts as a decision layer that combines and refines predictions of heterogeneous base models and further enhances predictive performance. To assure the validity and generalizability of the stacked ensemble, five-fold cross-validation tests and repeatedly divide the dataset into distinct training and validation sets. Of the total data, 70% was allocated for training and 30% for testing, allowing for a reliable evaluation with a separate test set. This prevents overfitting and guarantees consistent generalization of the final model on different data distribution. The stacked ensemble architecture harnesses the strengths of several machine-learning models, thereby enhancing robustness and predictive accuracy; hence it is a useful approach in the risk assessment of cardiovascular disease.

3.3.3. Evaluation metrics

Metrics like accuracy, precision, recall, F1-score, and ROC-AUC score help evaluate the model performance. These metrics shed light on how effectively a model distinguishes a person at risk of having cardiovascular disease from one who is not. The performance of each base learner is measured in isolation and compared to the stacked ensemble model, estimating improvements to the prediction accuracy owing to stacking. From the results, it is seen that the stacked ensemble always outperforms those based on a single learner, thereby substantiating this in its effectiveness for combining multiple learning strategies into one.

3.3.4. Model interpretability using explainable AI

To ensure transparency and trust in the predictions of a model made by AI, explainable AI (XAI) techniques, such as PFI and ALE, can be used. PFI measures the contribution of each feature to the predictions of the model by randomly permuting the values of that feature and estimating the drop in accuracy due to that perturbation [42]. This technique identifies the most important principal components in predicting cardiovascular diseases. ALE work to find the nonlinear interaction between predictors and model predictions [43]. Thus, by looking at the ALE values, researchers can infer how changes in individual features affected the model's decision-making process. The combination of PFI and ALE will ensure that the predictions can be interpreted and that the model can find application in real-world clinical settings.

4. Results

4.1. Dataset characteristics

The analysis included 4,240 records from the Framingham Heart Disease Dataset, with 3,596 instances in the majority class (no risk of cardiovascular disease) and 644 instances in the minority class (positive risk). The key attributes included demographic factors (age, gender), clinical indicators (systolic and diastolic blood pressure, glucose levels), and behavioral factors (smoking habits, BMI). The dataset exhibited a significant class imbalance, with only 15.19% of instances labeled as positive risk.

4.2. Model performance

4.2.1. Base models

Table 1 provides an overview of the performance outcomes for each individual models, highlighting their strengths and limitations in handling class imbalance. XGBoost achieved the highest accuracy (94.41%) with a precision of 91.69% and recall of 97.67%, while CatBoost demonstrated superior recall (98.39%) and precision (91.89%), resulting in an F1-score of 95.03%. LightGBM maintained a balanced performance with 93.29% accuracy and an F1-score of 93.57%. Despite their strengths, all base models struggled to balance precision and recall for the minority class, emphasizing the need for a more robust approach.

Table 1
Evaluation metrics for individual ML classifiers.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
XGBoost	94.41	91.69	97.67	94.59	99.07
LightGBM	93.29	89.73	97.75	93.57	98.64
CatBoost	94.85	91.89	98.39	95.03	99.18
Gradient Boosting Machine	82.11	76.64	92.36	83.77	90.57
AdaBoost	71.89	68.76	80.21	74.04	78.39
MLP without using Base Learners	85.50	83.12	88.45	85.71	78.5

4.2.2. Stacked ensemble model

The stacked ensemble model, integrating predictions from all base models with a MLP as the *meta*-model, outperformed individual base models across all evaluation metrics. Table 2 showcases the performance of the stacked model. It achieved an accuracy of 96.06%, a precision of 94.62%, a recall of 97.67%, and an F1-score of 96.12%, with the highest AUC-ROC of 99.31, indicating excellent discrimination between positive and negative classes. The high recall ensures that critical positive cases of CVD risk were accurately identified, minimizing false negatives. Fig. 2 shows the confusion matrix for the stacked model, demonstrating its superior predictive performance.

4.2.3. Class imbalance impact

The application of the SMOTE significantly improved the model's ability to detect minority class instances. As shown in Table 2, the recall for the minority class increased from 80.21% (AdaBoost) to 97.67% (Stacked Model), demonstrating the effectiveness of SMOTE in addressing class imbalance.

4.2.4. Feature importance

The PFI analysis depicted in Fig. 3 provides insights into the relative contribution of each principal component (PC) in predicting cardiovascular disease across various models. The results indicate that PC2 and PC6 consistently emerge as the most influential features across all models, suggesting that these principal components capture the most significant variance related to cardiovascular risk factors. Among the base models, CatBoost (PFI Score: 0.161 for PC6, 0.146 for PC2) and XGBoost (PFI Score: 0.153 for PC6, 0.136 for PC2) exhibit the highest sensitivity to these components. Similarly, LightGBM assigns significant importance to PC6 (PFI Score: 0.147) and PC2 (PFI Score: 0.122), reinforcing their impact on model predictions. Gradient Boosting and AdaBoost show lower overall feature importance scores, with PC6 and PC2 still ranking as top contributors, though with smaller magnitudes. The Stacked Model, which combines multiple classifiers, demonstrates the highest PFI scores overall, further confirming the dominance of PC6 (PFI Score: 0.184) and PC2 (PFI Score: 0.179). Additionally, PC3, PC9, and PC4 play a crucial role in shaping the final prediction, suggesting that they contain additional risk-related information. The superior feature attribution in the stacked model emphasizes its ability to extract and integrate key patterns from multiple base learners, enhancing cardiovascular disease prediction accuracy.

Fig. 4, the ALE plots illustrate the non-linear relationships between the principal components and model predictions. It is apparent that step-like patterns are present, especially for PC2, PC3, and PC6, indicating that the cardiovascular risk is influenced by these variables in a threshold-based manner. The razor-sharp changes seen in the ALE for PC2 suggest that certain levels beyond that threshold exert a strong influence on risk assessment by the model. Thus, it remains an important predictor in disease prediction. PC6 further reveals strong nonlinear effects with steep changes in effect values, thereby suggesting possible interaction effects of the different cardiovascular biomarkers. An opposite picture emerges for PC3 and PC9, showing more modest effects with relatively smooth ALE transitions, which suggests a more gradual way of affecting risk stratification. Thus, ALE supports the information obtained from PFI, confirming the importance of PC2, PC6, and PC3 and emphasizing their complex nonlinear effects on decisions made by the model. These interactions are well captured by the stacked model, thereby enhancing its interpretability and clinical relevance.

The PFI and ALE results demonstrate that PC2, PC6, and PC3 are the most influential components in predicting CVD risk, with nonlinear effects shaping their impact. The stacked model outperforms individual base learners, achieving higher feature importance scores and capturing stronger nonlinear interactions. These findings enhance the interpretability of AI-driven cardiovascular prediction models, making them more trustworthy and suitable for clinical decision-making.

Table 2
Evaluation metrics of the stacked ensemble model for CVD prediction.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Stacked Model	96.06	94.62	97.67	96.12	99.31

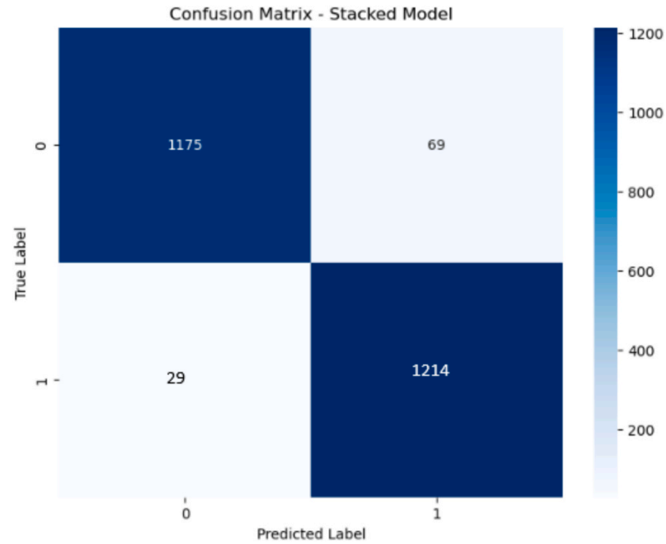

Fig. 2. Confusion Matrix of the Stacked Ensemble Model.

4.2.5. Feature importance correlation analysis across models

The Feature Importance Correlation Matrix in Table 3 and the corresponding heatmap in Fig. 5 illustrate the degree of agreement among different machine learning models in ranking feature importance for cardiovascular disease prediction. The correlation values indicate how consistently each model identifies significant predictors, with higher values suggesting strong alignment in feature selection. The results reveal that most models exhibit high correlation, with coefficients exceeding 0.80, highlighting strong consistency in feature attribution across different machine learning approaches.

The analysis shows that Gradient Boosting and AdaBoost share the highest correlation (0.96) among base learners, indicating that both models emphasize similar principal components in their decision-making process. This strong alignment suggests that boosting-based approaches, which iteratively refine weak learners, tend to identify comparable patterns in feature importance. AdaBoost's reliance on adjusting misclassified instances aligns well with Gradient Boosting's method of sequential learning, leading to a high degree of similarity in their feature ranking.

The Stacked Model demonstrates a near-perfect correlation with CatBoost (0.991) and a strong correlation with LightGBM (0.9525), suggesting that these two models significantly influence the final predictions of the stacked ensemble. Since CatBoost is optimized for handling categorical variables and LightGBM is designed for computational efficiency, their strong correlation with the stacked model indicates their substantial contribution to the ensemble's decision-making process. The stacked model effectively integrates multiple learning algorithms, utilizing the strengths of these base models to improve cardiovascular disease prediction accuracy.

Another interesting finding is that the correlation between XGBoost and LightGBM is very high (0.9478). This close relationship is explained by the fact that both of them follow a gradient boosting architecture that allows the robust use of structured data and captures complex feature interactions. Due to this similarity in the optimization and handling of large datasets, XGBoost and LightGBM are remarkably aligned in regard to their feature importance rankings.

However, when comparing AdaBoost and XGBoost, the correlation is

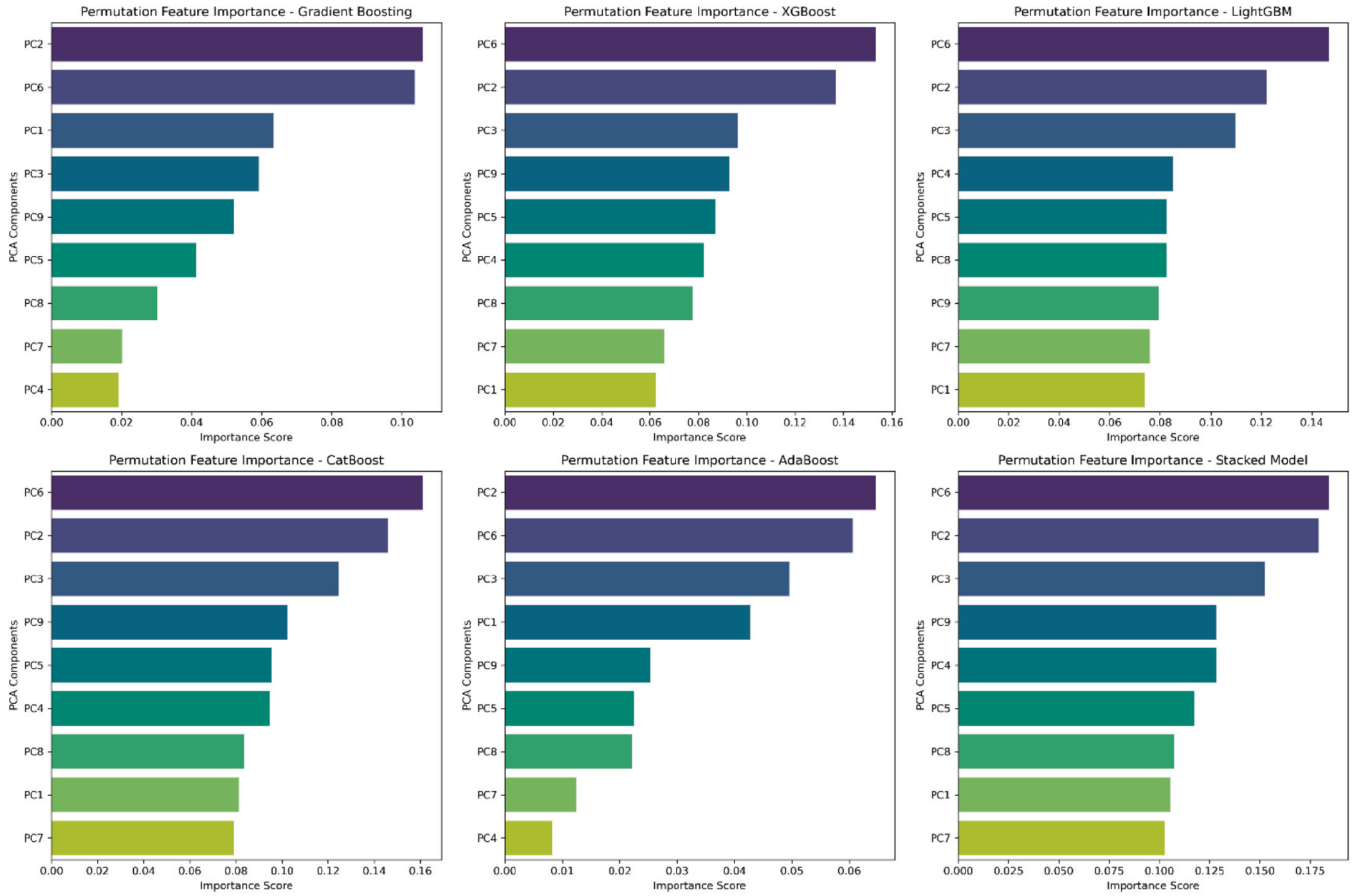

Fig. 3. Permutation Feature Importance (PFI) scores across different machine learning models.

considered quite low with scores of about 0.7528 compared to the other combinations. This proves that AdaBoost constitutes a different mechanism of weighting for importance of the feature, focusing on peculiar features of the data not necessarily present in XGBoost. The fact that they belong to the boosting family does not prevent the model's emphasis from being on re-weighting samples misclassified in each iteration, which can in turn lead to slight discrepancies in their importance of features when compared with XGBoost's gradient-based optimization.

Again, supporting these facts is a heat map. The darker the red areas, the higher the amount of agreement between the model samples; conversely, the cooler the blue shades, the lower the amount of agreement. This strong amount of agreement between the Stacker Model and CatBoost is also emphasized in the heat map. It reaffirms the finding that the features identified and selected by CatBoost tend to play sensational roles in determining the final ensemble predictions. Moreover, the closeness between Gradient Boosting and AdaBoost is also denoted graphically and strengthens the basis for allowing shared feature selection. The comparatively lesser amount of correlation between AdaBoost and XGBoost is also evident, emphasizing the use of feature weighting.

All in all, high correlation among models essentially serves to confirm stability and reliability in terms of feature selection so that important principal components will consistently get to be viewed as significant predictors of cardiovascular disease risk. In summary, the Stacked Model is able to learn from patterns of several base learners and present the characteristics that CatBoost and LightGBM contribute extensively to its final predictions. The importance of ensemble learning is further stressed in terms of harnessing robust feature importance patterns shown by this heat map analysis. This strengthens the model's interpretability and provides confidence in the AI-based frameworks for cardiovascular disease prediction.

4.2.6. *Comparative analysis of model similarity and correlation*

The Jaccard Similarity, Pearson Correlation, Spearman Correlation, and Kendall Correlation Matrices provide insights into the relationships among different ML models in predicting CVD risk. These matrices measure how similarly models rank feature importance and make predictions, helping assess the consistency and complementarity of different learning approaches.

The Jaccard Similarity Matrix in Table 4 quantifies the overlap between models in terms of selected important features. The results indicate a high similarity between XGBoost and LightGBM (0.9321), XGBoost and CatBoost (0.9450), and CatBoost and LightGBM (0.9372), reinforcing the alignment of these gradient boosting-based models in identifying key risk factors. Additionally, the Stacked Model shows strong similarity with CatBoost (0.9624) and XGBoost (0.9528), suggesting that these models significantly contribute to the final ensemble decision-making process. However, AdaBoost demonstrates the lowest Jaccard similarity with other models, particularly with XGBoost (0.6279) and CatBoost (0.6215), indicating that it selects features differently from the other models.

The Pearson Correlation Matrix in Table 5 measures the linear relationships between model outputs. The highest correlations appear between XGBoost and LightGBM (0.9242), XGBoost and CatBoost (0.9394), and CatBoost and the Stacked Model (0.9604). These findings suggest that models based on gradient boosting tend to assign similar importance to key features, reinforcing their consistency in prediction. AdaBoost shows lower Pearson correlation scores, particularly with CatBoost (0.4741) and the Stacked Model (0.4601), indicating that its

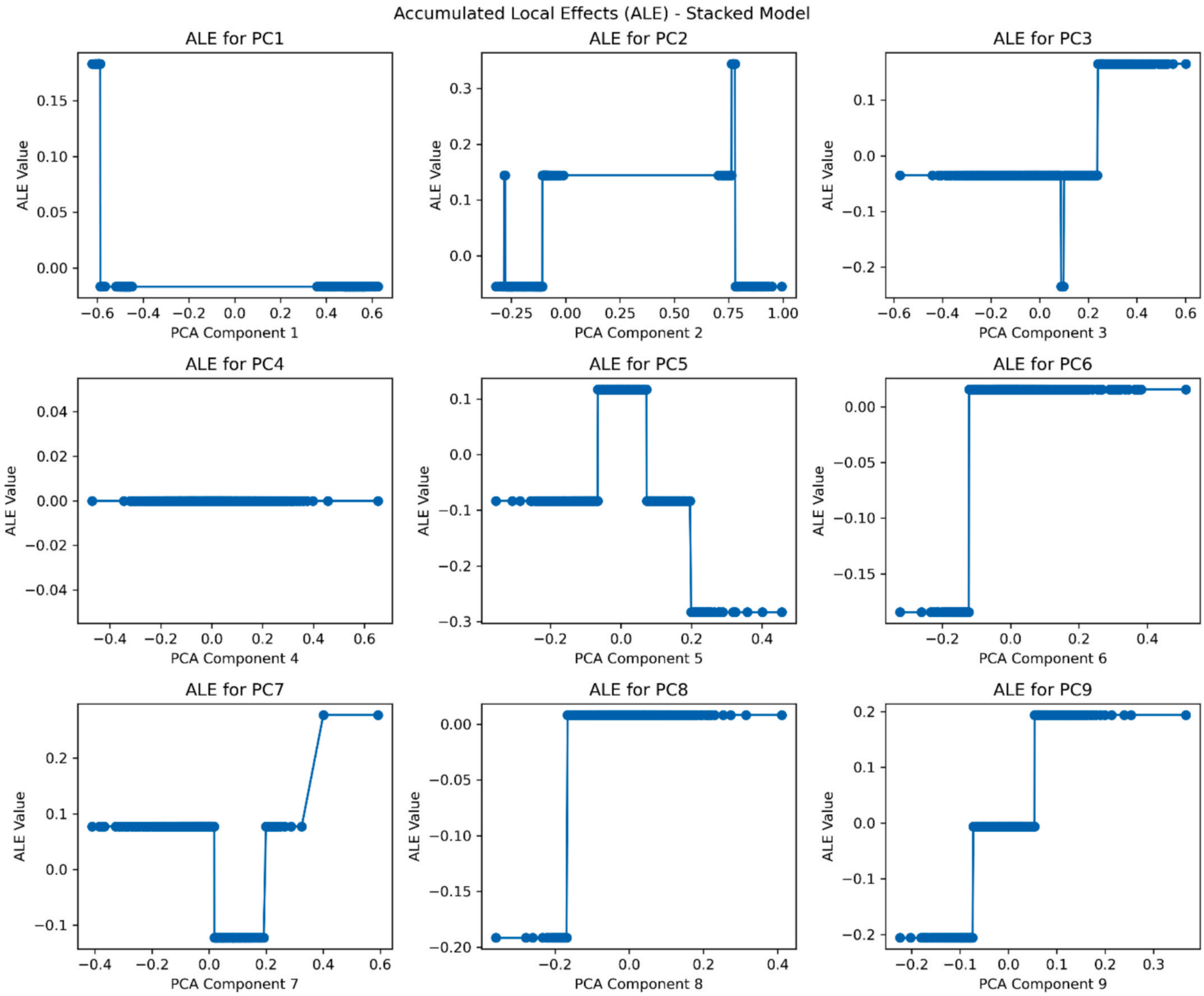

Fig. 4. Accumulated Local Effects (ALE) plots for principal components influencing cardiovascular disease risk.

Table 3
Feature Importance Correlation Matrix across different machine learning models.

Model	AdaBoost	CatBoost	Gradient Boosting	LightGBM	Stacked Model	XGBoost
AdaBoost	1.00000	0.82268	0.95974	0.79782	0.80157	0.75279
CatBoost	0.82268	1.00000	0.86592	0.97115	0.99144	0.97414
Gradient Boosting	0.95974	0.86592	1.00000	0.8158	0.84371	0.84405
LightGBM	0.79782	0.97115	0.8158	1.00000	0.95255	0.94782
Stacked Model	0.80157	0.99144	0.84371	0.95255	1.00000	0.96012
XGBoost	0.75279	0.97414	0.84405	0.94782	0.96012	1.00000

feature weighting mechanism differs significantly from the gradient boosting methods.

The Spearman Correlation Matrix in Table 6, which evaluates monotonic relationships, displays similar trends to the Pearson correlation results. The Stacked Model and CatBoost exhibit the highest correlation (0.9604), followed by XGBoost and LightGBM (0.9242). These strong correlations further confirm that models utilizing boosting techniques assign importance to features in a similar manner. AdaBoost once again shows the lowest correlation scores, reinforcing its distinct feature selection and prediction strategy.

The Kendall Correlation Matrix in Table 7, which measures ordinal associations, also reflects strong relationships among gradient boosting models, particularly between XGBoost, LightGBM, and CatBoost. The Stacked Model maintains strong correlations with CatBoost (0.9604) and XGBoost (0.9497), reinforcing the importance of these models in shaping its final predictions. AdaBoost continues to show weaker correlations, particularly with CatBoost (0.4741) and the Stacked Model (0.4601), suggesting that it captures different aspects of feature importance compared to the other models.

Fig. 6 illustrates the Model Similarity and Correlation analysis across various machine learning approaches. The high correlation scores between XGBoost, LightGBM, and CatBoost across all matrices indicate that these models consistently identify similar feature importance patterns and make predictions in a closely aligned manner. The Stacked

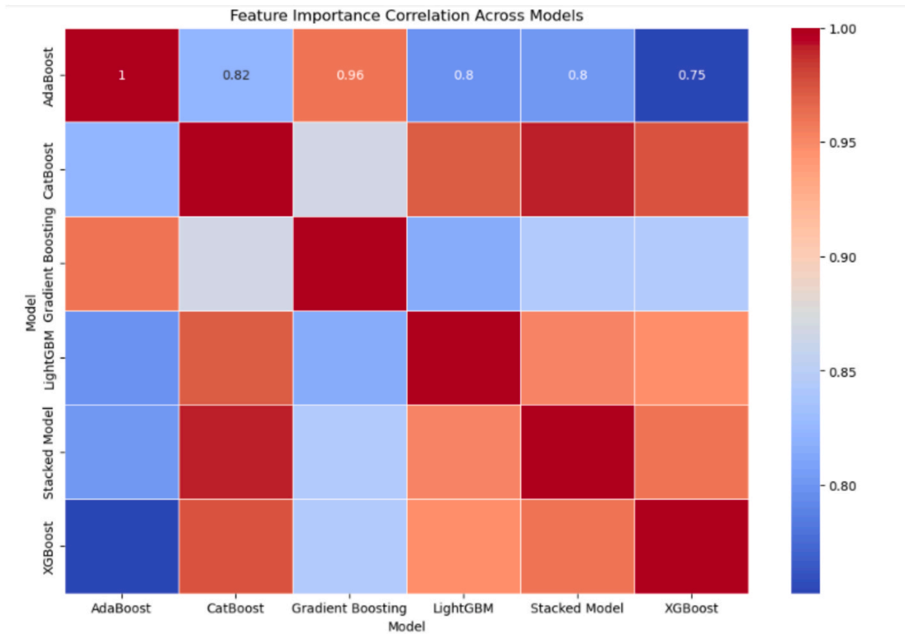

Fig. 5. Feature Importance Correlation Heatmap highlighting agreement among different models.

Table 4
Jaccard Similarity Matrix of machine learning models based on feature selection.

	Gradient Boosting	XGBoost	LightGBM	CatBoost	AdaBoost	Stacked Model
Gradient Boosting	1.000000	0.781566	0.799369	0.777010	0.746445	0.755682
XGBoost	0.781566	1.000000	0.932179	0.945055	0.627934	0.952809
LightGBM	0.799369	0.932179	1.000000	0.937229	0.637850	0.926224
CatBoost	0.777010	0.945055	0.937229	1.000000	0.621574	0.962462
AdaBoost	0.746445	0.627934	0.637850	0.621574	1.000000	0.604815
Stacked Model	0.755682	0.952809	0.926224	0.962462	0.604815	1.000000

Table 5
Pearson Correlation Matrix illustrating linear relationships between model outputs.

	Gradient Boosting	XGBoost	LightGBM	CatBoost	AdaBoost	Stacked Modell
Gradient Boosting	1.000000	0.725347	0.744731	0.717032	0.644215	0.697387
XGBoost	0.725347	1.000000	0.924242	0.939425	0.487170	0.949751
LightGBM	0.744731	0.924242	1.000000	0.929757	0.495463	0.920057
CatBoost	0.717032	0.939425	0.929757	1.000000	0.474143	0.960420
AdaBoost	0.644215	0.487170	0.495463	0.474143	1.000000	0.460128
Stacked Model	0.697387	0.949751	0.920057	0.960420	0.460128	1.000000

Table 6
Spearman Correlation Matrix depicting monotonic relationships among models.

	Gradient Boosting	XGBoost	LightGBM	CatBoost	AdaBoost	Stacked Model
Gradient Boosting	1.000000	0.725347	0.744731	0.717032	0.644215	0.697387
XGBoost	0.725347	1.000000	0.924242	0.939425	0.487170	0.949751
LightGBM	0.744731	0.924242	1.000000	0.929757	0.495463	0.920057
CatBoost	0.717032	0.939425	0.929757	1.000000	0.474143	0.960420
AdaBoost	0.644215	0.487170	0.495463	0.474143	1.000000	0.460128
Stacked Model	0.697387	0.949751	0.920057	0.960420	0.460128	1.000000

Model exhibits the highest similarity with CatBoost and XGBoost, emphasizing their contribution to its overall predictive power. On the other hand, AdaBoost shows the lowest similarity and correlation with the other models, highlighting its unique feature weighting approach. These findings confirm that ensemble learning effectively integrates complementary models, with the Stacked Model leveraging the strengths of gradient boosting-based approaches while balancing the differences introduced by AdaBoost. The results further validate the robustness and stability of the selected feature importance rankings, ensuring a high degree of reliability in cardiovascular disease prediction.

4.2.7. Comparative analysis

The ROC curve shown in Fig. 7 expresses the performance of various ML models, GBM, XGBoost, CatBoost, LightGBM, AdaBoost, and Stacked Model, in predicting CVD. The AUC values under the curve represent the

Table 7
Kendall Correlation Matrix showing ordinal associations between machine learning models.

	Gradient Boosting	XGBoost	LightGBM	CatBoost	AdaBoost	Stacked Modelll
Gradient Boosting	1.000000	0.725347	0.744731	0.717032	0.644215	0.697387
XGBoost	0.725347	1.000000	0.924242	0.939425	0.487170	0.949751
LightGBM	0.744731	0.924242	1.000000	0.929757	0.495463	0.920057
CatBoost	0.717032	0.939425	0.929757	1.000000	0.474143	0.960420
AdaBoost	0.644215	0.487170	0.495463	0.474143	1.000000	0.460128
Stacked Model	0.697387	0.949751	0.920057	0.960420	0.460128	1.000000

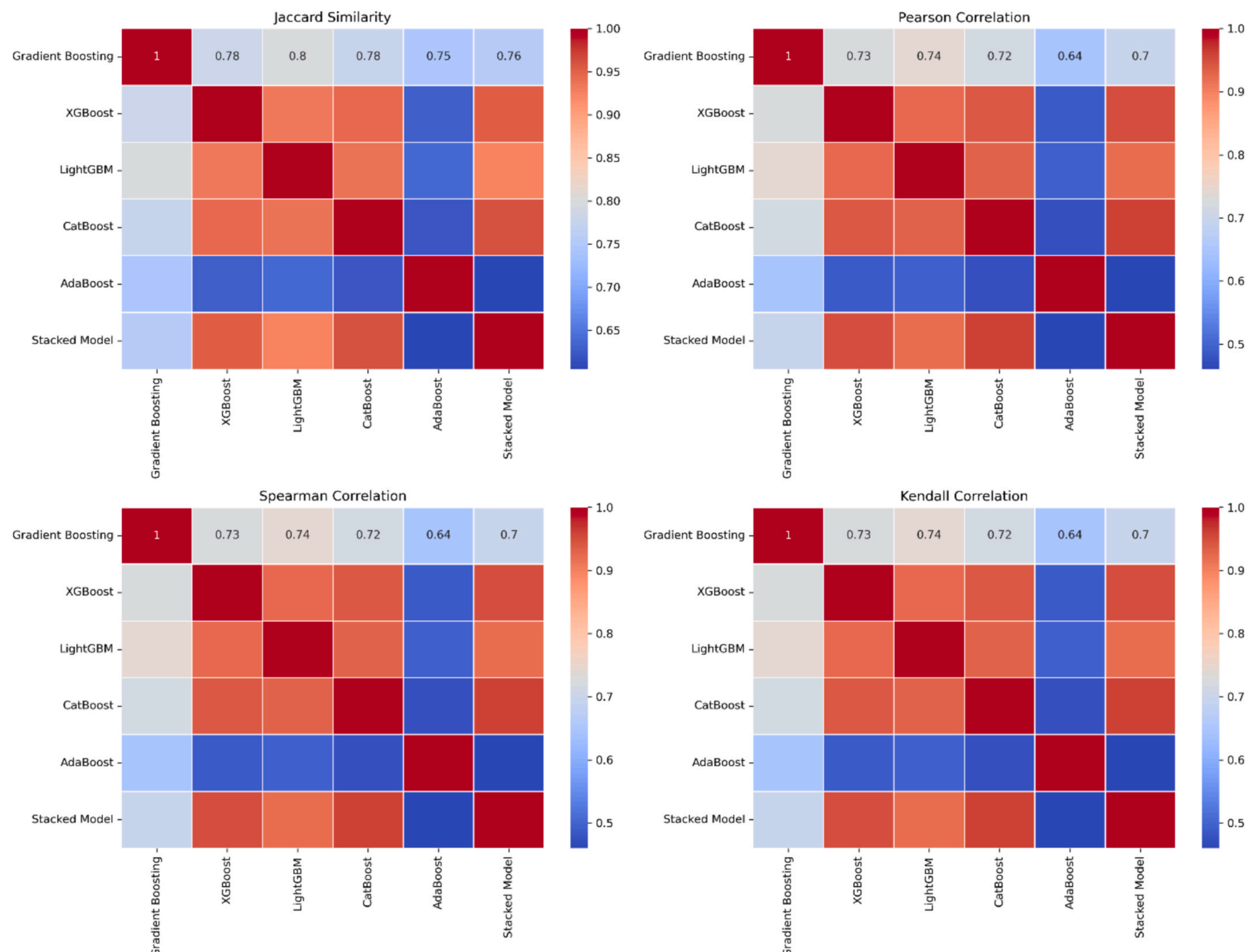

Fig. 6. Model Similarity and Correlation analysis across different machine learning approaches.

predictive power of the models with higher values showing how well it can differentiate a positive from a negative case. The result showed that among the base learners, XGBoost AUC (0.9907), LightGBM AUC (0.9864), and CatBoost AUC (0.9818) scored best in prediction, while Gradient Boosting AUC (0.9057) and AdaBoost AUC (0.7839) scored lower. The best AUC of 0.9931 went to the stacked model, which outperforms all standalone base models. This result highlights the effectiveness of stacked ensemble learning, which uses the strengths of multiple classifiers to enhance overall predictive accuracy and robustness. The stacked model's superior performance confirms that combining diverse learning algorithms enhances generalization and reliability, making it the most effective approach for cardiovascular disease prediction.

5. Discussion

5.1. Principal findings

This work proposed a stacked ensemble learning strategy to predict CVD risk, utilizing the Framingham dataset. The preprocessing phase involved managing missing entries, addressing class imbalance, and applying feature normalization to ensure data quality and consistency throughout model development. The ensemble framework incorporated XGBoost, LightGBM, CatBoost, Gradient Boosting, and AdaBoost, with a MLP acting as the *meta*-learner. This configuration achieved a top predictive accuracy 96.06% and an AUC-ROC score 99.31, reflecting outstanding model performance.

Utilizing SMOTE led to a notable enhancement in model

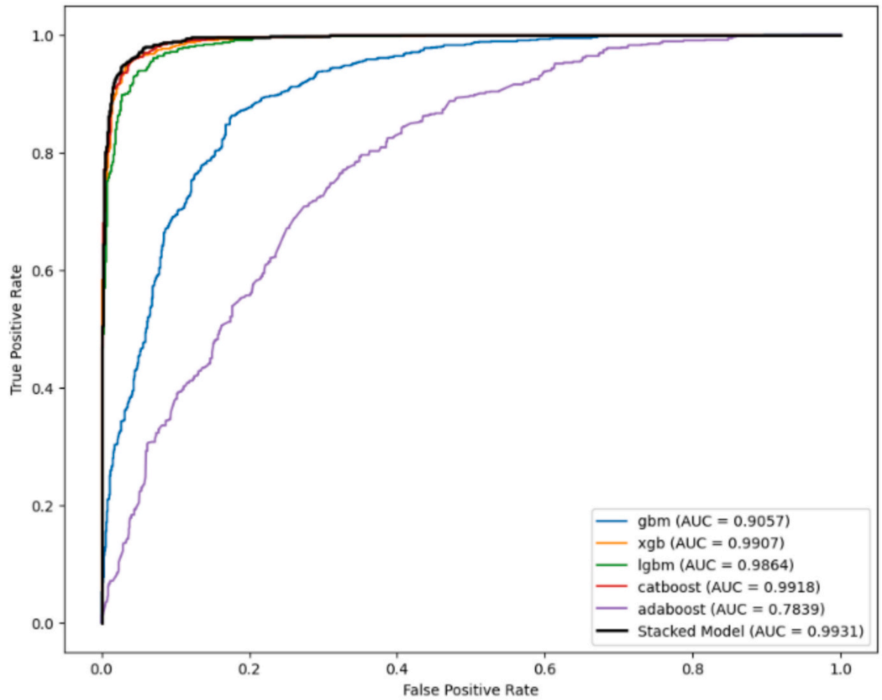

Fig. 7. ROC curve comparing the classification performance of different models.

performance by addressing class imbalance, thus minimizing bias in favor of the dominant class. Through PFI and ALE analysis, the study identified PC2 (linked to hypertension and blood pressure), PC6 (related to smoking and cholesterol), and PC3 (involving age and cholesterol) as the most critical variables influencing cardiovascular disease (CVD) risk. These results emphasize the value of ensemble learning approaches, which integrate various machine learning algorithms to boost both predictive performance and clinical applicability.

5.2. Clinical Implications

This study's findings have enormous clinical relevance for improving CVD risk prediction and preventive healthcare. The very high recall (97.67%) in the stacked ensemble model means very few high-risk patients have been missed clinically. Thus, early interventions may be instituted once CVD screening provides a positive signal. The model complements risk stratification with the identification of key risk factors for intervention, like hypertension, smoking, and cholesterol levels, to inform and provide feedback about possible interventions and lifestyle changes. The PFI and ALE analyses enhance transparency; therefore, clinicians understand how specific features affect predictions, and this builds confidence in using this AI for diagnosis. The model, with high accuracy and interpretability, would be easily embedded into EHR systems for the use of healthcare professionals in making objective personalized treatment decisions that will increase patient outcomes and resource allocation in clinical settings.

5.3. Addressing class imbalance

A significant obstacle in clinical datasets is the natural class imbalance, which often results in skewed predictions that disproportionately favor the majority class. By employing SMOTE, this study demonstrated a significant improvement in recall for the minority class (from 80.21% in AdaBoost to 97.67% in the stacked model). This ensures that critical cases of CVD risk are not overlooked, addressing a key limitation of conventional models.

5.4. Comparison with previous studies

The stacked ensemble model achieves an accuracy of 96.06%, significantly outperforming previous studies in cardiovascular disease (CVD) risk prediction. By integrating Gradient Boosting Machine (GBM), XGBoost, LightGBM, CatBoost, and AdaBoost as base learners, with a MLP *meta*-learner, this model enhances predictive performance while ensuring class balancing and interpretability. These enhancements position it as a promising solution for real-world clinical applications.

In contrast, prior research has reported lower accuracy levels using various machine learning approaches. For instance, Sudipta et al. [44] achieved 87.70% accuracy using a Multilayer Perceptron (MLP) across various datasets, including Cleveland, Hungarian, Switzerland, Long Beach, and StatLog, applying infinite feature selection techniques [44]. Similarly, Reddy et al. [45] used Sequential Minimal Optimization (SMO) on the Cleveland Heart Dataset, reaching 85.15% accuracy with the full attribute set and 86.47% with an optimized attribute selection [45].

Other studies employed different machine learning techniques but still recorded lower predictive performance. Recently, a Hybrid Random Forest with Linear Model was proposed on the UCI Cleveland dataset, achieving an accuracy of 88.7% using 13 clinical features [46]. Ambrews et al. [47] improved performance slightly with a Voting Ensemble Model applied to the UCI dataset, obtaining 91.96% accuracy [47]. Another study employed a stacked ensemble model with ten base learners, including Random Forest (RF), MLP, KNN, Extra Trees (ET), XGBoost, Support Vector Classifier, Stochastic Gradient Descent, AdaBoost (ADB), CART, and Gradient Boosting Machine (GBM), and reached 92% accuracy [41].

Compared to the earlier results, the new stacked ensemble model achieves really high benchmark accuracy with significant prediction ability of 96.06%. Sophisticated GBM algorithms with a strong neural network *meta*-learner improved risk stratifications among CVD patients compared to other works mentioned in the literature. Other than high accuracy, the explainability of the model has been a concern, utilized via explainable AI (XAI) methods such as PFI and ALE. Therefore, the superior performance in CVD risk prediction, class balancing, and interpretability by this new stacked model was clearly established in comparison with previous research. The accuracy was 96.06%, thereby beating the previous ones with all the machine learning methods needed to provide transparency through-and-through XAI methods. All this renders the model a viable early predictor of diagnosis and risk stratification for age-old competition, enhancing cardiovascular health outcomes through personalized patient-centered care at the bedside.

5.5. Feature importance and interpretability

To enhance interpretability and transparency, methods of explainable AI such as PFI and ALE were exploited. The analysis identified PC2 (Hypertension & Blood Pressure) to be the most important as it strongly correlates with systolic and diastolic blood pressure levels and widespread hypertension, thus presenting its direct impact on CVD risk. On the other hand, PC6 (Smoking & Cholesterol) highlighted destructive effects of cigarette smoking and high cholesterol levels on cardiovascular health, while PC3 (Age & Cholesterol) showed age-related change in lipid metabolism, highlighting its importance in the risk estimate. The nonlinear effects shown in PC2 and PC6 through the ALE analysis revealed that the respective threshold levels of blood pressure and smoking have critical significance in the map of CVD risk estimation. This finding gives clinically interpretable results, therefore empowering medical professionals to make patient-specific patient management decisions and thereby optimizing preventive care interventions and treatment outcomes.

5.6. Future Directions

Future research should validate the model on multi-ethnic datasets for better generalizability and incorporate genetic, lifestyle, and socioeconomic factors to enhance predictive power. Deep learning comparisons may reveal longitudinal health patterns, while hyperparameter tuning could further optimize performance. Clinical testing and hospital integration will assess real-world feasibility, and a dynamic risk prediction approach could enable continuous monitoring and early interventions, improving preventive care and treatment strategies.

6. Conclusion

This research successfully developed a stacked ensemble learning model for predicting CVD risk, achieving high predictive accuracy (96.06%) and AUC-ROC (99.31%). The model effectively addressed class imbalance using SMOTE, improved dimensionality reduction with PCA, and ensured interpretability through Explainable AI (XAI) techniques like PFI and ALE. PFI analysis identified the most important components with strong effects on CVD risk as PC2 (hypertension and blood pressure), PC6 (smoking and cholesterol), and PC3 (age and cholesterol). ALE analysis also identified nonlinear effects in key variables with threshold-based effects on CVD predictions and with higher clinical interpretability. The results enhance the capacity of the model to personalize risk stratification and enable data-driven interventions. In spite of robust model performance, future research should strive to validate the model in different populations, genetic, lifestyle, and socioeconomic inputs, and improvement in performance via hyperparameter tuning. Real-world clinical trials and dynamic risk prediction approaches can further enhance patient monitoring and early interventions. Leveraging XAI techniques like PFI and ALE, the study spotlights the importance of explanation and trust in AI models, and the framework thus becomes an explainable and scalable solution for improving CVD risk estimation, clinical decision-making, and preventive healthcare planning.

Ethics statement

This study does not require ethical approval as it is based on the publicly available and anonymized Framingham Heart Disease dataset, with no direct human or animal involvement, ensuring compliance with relevant institutional and ethical guidelines.

CRediT authorship contribution statement

Jeena Joseph: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **K. Kartheeban:** Conceptualization, Methodology, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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